

BCA (Semester-II) Examination, 2022

(Session 2020-23)

COMPUTER APPLICATION

| Paper Code : BCA-201 |

(Discrete Mathematics)

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Define the following terms:

- (a) Inclusion and Exclusion of set
- (b) Ven diagram
- (c) Equality of sets
- (d) Rings

2. If $A = \{1, 2, 3\}$, $B = \{2, 4, 5\}$ and $\Omega C = \{x : x \text{ is a digit}\}$, then:

- (a) $(A \cup B) \cap (B - A)$
- (b) $(A - B)^c$
- (c) $A^c - B^c$
- (d) $(A \cap C)^c$

3. (a) Define equivalence relation on a set with example.

(b) If $A = \{1, 2, 3, 4\}$, $B = \{1, 3, 9, 10\}$, $C = \{5, 6, 7, 8\}$ and $R = \{(1, 1), (1, 3), (2, 9), (2, 10), (3, 3), (4, 10)\}$ and $S = \{(1, 5), (3, 7), (9, 7), (10, 8)\}$, then find R o S and its relation graph.

4. List all partition of sets:

(a) $A = \{x, y, z\}$

(b) $B = \{1, 2, 3, 4\}$

5. If the function $f : R \rightarrow R$ be defined by $f(x) = x^2 + 1$, where R is the set of real numbers, then find the value of:

(a) $f^{-1}(-5)$

(b) $f^{-1}(26)$

(c) $f^{-1}(10, 37)$

6. If $X = \{2, 3, 6, 12, 24, 36\}$, R on $X = \{(x, y) \in R \mid x \text{ divides } y\}$.

(a) Construct Hasse diagram

(b) Maximal and Minimal elements

(c) Is POSET a lattice

7. (a) Draw a K-map 4-variable.

(b) If $f(xs) = 3x - 5$ and $f[g(x)] = 2x$ then find $g(x)$

8. Show the set of all divisors of 70 from a Lattice.

9. Minimize the Boolean expression using K-map:

(a) $f = x'yz + xy'z + xyz + xyz'$

(b) $f = A'C + A'B + AB'C + BC$

10. Write short notes on any two of the following:

(a) Equivalence Relation

(b) Monoid

(c) Semi-group

(d) Partitions of set

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Grab Notes